

Timothy Yang

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EDUCATION

Southern Connecticut State University

Master of Science in Computer Science, GPA 3.5

New Haven, CT

Jan. 2026 – Dec. 2027

Southern Connecticut State University

Bachelor of Science in Computer Science, Minor in Mathematics, GPA 3.47

New Haven, CT

Jan. 2024 – Dec. 2025

WORK EXPERIENCE

AV/IT Technician

Southern Connecticut State University

Apr. 2026 – Present

New Haven, CT

- Collaborated with IT staff on AV/TV troubleshooting, maintenance, and documentation, reducing incidents.
- Created a dynamic visual dashboard using Power BI and VBScript to monitor classroom and student usage hours.

Information Technology Intern

Connex Credit Union

May 2023 – Aug. 2023

North Haven, CT

- Supported a core-system migration by validating data integrity and records for accurate post-conversion data.
- Resolved multiple daily helpdesk tickets across hardware, software, and network issues for 100+ employees.

PROJECTS

Distributed Trading Card Tracker | *Java, Kotlin, Spring, Kafka, Cassandra, MySQL/Postgres* Mar. 2026 – Pres.

- Architected a distributed, event-driven TCG collection manager (CQRS) in Java 23, Kotlin, and Spring Boot 4, separating the MySQL/JPA write path from scalable Cassandra read models over an Apache Kafka event bus.
- Migrated the read-side event-projection path (Kafka event data classes and the Kafka→Cassandra projector) to Kotlin within a mixed Java/Kotlin/Groovy Spring Boot module, preserving the snake_case Kafka wire contract.
- Tested the CQRS core with Spock/Groovy and JUnit specs covering write-side persistence, event publishing, Kafka→Cassandra projection routing, and a @WebMvcTest slice asserting JWT-protected endpoint behavior.
- Instrumented the service with Micrometer → InfluxDB and Grafana, exposing command and projection counters plus price-fetch and catalog-search latency timers (p95) tagged by service role on a Docker-verified dashboard.
- Built a polyglot-persistence backend across MySQL (JPA), Supabase Postgres (catalog search over 7K+ cards), and Cassandra, using denormalized read models for JOIN-free, single-partition lookups.

AI Speedcubing Assistant | *Python, Flask, Claude, pgvector, TiDB, Docker, Fly.io*

Jan. 2026 – Pres.

- Built a natural-language-to-SQL (NLP) interface over a 6.3M+ row WCA competition database, using Claude API to translate plain-English questions into validated, read-only SQL for non-technical users.
- Developed a Retrieval-Augmented Generation (RAG) chatbot for WCA regulations with Voyage AI embeddings, pgvector similarity search, and a reranker over 697 regulations, returning grounded answers with citations.
- Reduced database maintenance from a 4-step process to a single command via an automated pipeline that fetches the WCA export API, streams and parses multi-GB TSV/SQL archives, and bulk-loads all tables into TiDB.
- Hardened the app with a strict CSP, per-endpoint rate limiting, and non-SELECT query rejection; deployed a multi-process Docker container (web app + Discord bot) to Fly.io via a GitHub Actions CI/CD pipeline.

Interactive Portfolio with CMS | *Next.js, React, TypeScript, Prisma, Turso, Claude*

May 2026 – Pres.

- Built a full-stack portfolio and session-authenticated headless CMS with Next.js 16, React 19, and TypeScript, deployed on Vercel with a Turso (SQLite/libSQL) database via the Prisma driver adapter.
- Designed an interactive, themeable UI in Framer Motion featuring a self-scrambling-and-solving CSS-3D Rubik's cube driven by rotation-matrix layer turns, cursor-reactive parallax geometry, and 3D-tilt project cards.
- Integrated the Anthropic Claude API to auto-draft project entries from GitHub READMEs and rewrite content from natural-language instructions, using prompt caching and adaptive thinking.
- Built a custom visitor-analytics pipeline (page views, downloads, messages) with geo detection in a Recharts dashboard, plus admin CRUD over all site content with no redeloys.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, TypeScript, JavaScript, SQL, Groovy, C, C++, C#, HTML, CSS

Frameworks & Tools: React, Next.js, Spring, Flask, Node, npm, Vite, Docker, Apache Kafka, JUnit, Spock, Git

Data & Cloud: PostgreSQL, MySQL, Cassandra, InfluxDB, Supabase, TiDB, Grafana, AWS, Fly.io, Vercel, GCP